Week 2 – Quantum Computing

Complex Numbers, Linear Algebra

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Review from Week 1

- Last week, we examined elementary examples of classical systems, both deterministic and probabilistic. In each case, the state of the system was represented by a vector, and time evolution was described by multiplication by a matrix.
- Deterministic systems were modeled by matrices with exactly one nonzero entry in each column, while probabilistic systems were modeled by column-stochastic matrices that preserve total probability.
- We saw that certain physical phenomena cannot be consistently modeled within this classical probabilistic framework. This motivates replacing probability vectors with complex vectors, and classical transition matrices with linear transformations that preserve a different quantity.
- This week, we begin developing the mathematical structure of quantum mechanics: states represented by normalized vectors in complex vector spaces (qubits), and time evolution described by so-called **unitary** matrices.

Matrix algebra preliminaries

Throughout, let A be an $n \times n$ matrix with complex entries.

Transpose. The *transpose* of A, denoted A^T, is obtained by swapping rows and columns:

$$(A^T)_{i,j} = A_{j,i}.$$

Entrywise complex conjugate. The *complex conjugate* of A, denoted A^*, is obtained by taking complex conjugates entry-by-entry:

$$(A^*)_{i,j} = (A_{i,j})^*.$$

Conjugate transpose. The *conjugate transpose* of A, denoted $A^\dagger$, is defined by

$$A^\dagger := (A^*)^T = (A^T)^*, \qquad \text{i.e.} \qquad (A^\dagger)_{i,j} = (A_{j,i})^*.$$

Example. Let

$$A = \begin{pmatrix} 1+i & 2-3i \\ -1+2i & 4 \end{pmatrix}.$$

Then

$$A^T = \begin{pmatrix} 1+i & -1+2i \\ 2-3i & 4 \end{pmatrix}, \qquad A^* = \begin{pmatrix} 1-i & 2+3i \\ -1-2i & 4 \end{pmatrix},$$

and

$$A^\dagger = \begin{pmatrix} 1-i & -1-2i \\ 2+3i & 4 \end{pmatrix}.$$

Hermitian and unitary matrices

Definition 1 (Hermitian)**.** *A square matrix H is called* Hermitian *if*

$$H^\dagger = H.$$

Example (Hermitian). Let

$$H = \begin{pmatrix} 2 & 1+i \\ 1-i & 3 \end{pmatrix}.$$

Compute

$$H^\dagger = \begin{pmatrix} 2^* & (1-i)^* \\ (1+i)^* & 3^* \end{pmatrix}^T = \begin{pmatrix} 2 & 1+i \\ 1-i & 3 \end{pmatrix} = H,$$

so H is Hermitian.

Definition 2 (Unitary)**.** *A square matrix U is called* unitary *if*

$$U^\dagger U = I \qquad \textit{(equivalently, } U^{-1} = U^\dagger\textit{)}.$$

Example (Unitary). Let

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Since U has real entries, $U^\dagger = U^T = U$. Then

$$U^\dagger U = U^T U = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = I.$$

Thus U is unitary, and indeed $U^{-1} = U^\dagger = U$.

Exercise 1. *Give an example of a 2×2 complex matrix that is:*

(a) both Hermitian and unitary;

(b) Hermitian but not unitary;

(c) unitary but not Hermitian.

First postulate – Introduction to the qubit

There are four central postulates of quantum mechanics that underlie quantum computing, that touch upon how quantum information is represented, how it is transformed, how it is measured, and how it is combined. We begin with the first one, which concerns **representation**.

First Postulate. Associated to any isolated physical system is a complex inner product space (called a Hilbert space), known as the state space of the system. The system is described by its state vector, which is a **unit** vector in the state space.

In general, a state vector is a (complex-valued) ket

$$|\psi\rangle = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix},$$

with $|c_1|^2 + \ldots + |c_n|^2 = 1$. The complex numbers c_i are called **amplitudes**, and the normalization condition ensures that the total probability of all outcomes equals 1.

In quantum computing and information, the simplest physical system / unit of information is called a **qubit** (quantum bit), and is represented by

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \qquad \alpha, \beta \in \mathbb{C}, |\alpha|^2 + |\beta|^2 = 1.$$

Using a standard orthonormal basis with $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, and $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, it is common to express a qubit as $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. We say that $|\psi\rangle$ is in a **superposition** of the basis states $|0\rangle$ and $|1\rangle$. Note that a qubit can be represented in *any* orthonormal basis $\{|\Phi_0\rangle, |\Phi_1\rangle\}$ for $\mathbb{C}^2$, and this is a notion that will be frequently useful in applications.

We remark at this time that multiplication of a qubit state by any complex number of unit modulus – called a "global phase" – represents **the same physical state**. For example, the ket $|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$ is to be regarded as **physically indistinguishable** from, say the ket $-|\psi\rangle$, or $i|\psi\rangle$, or indeed any $e^{i\phi}|\psi\rangle$. *Relative* phase, on the other hand – consider $|\psi'\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$ – **is** significant and is to be regarded as a distinct physical qubit state. We will explore these concepts in greater detail soon.

Second Postulate – unitary evolution

Second Postulate. The evolution of a closed quantum system is described by a unitary transformation. In other words, a state $|\psi\rangle$ at time t_1 relates to $|\psi'\rangle$ at time t_2 by $|\psi'\rangle = U|\psi\rangle$, where U is a unitary operator.

In the context of quantum computing, what this means is that qubits can be manipulated and transformed by unitary matrices called as **quantum gates**.

Exercise 2. *The following problems highlight key structural properties of unitary matrices. Recall that*

$$\| \, |x\rangle \, \| := \sqrt{\langle x|x\rangle}.$$

1. *Prove that if U and V are unitary matrices, then so is UV.*

2. *Prove that any unitary matrix U is norm-preserving: for any ket $|x\rangle$,*

$$\|U|x\rangle\| = \||x\rangle\|.$$

3. *Prove that any unitary matrix U is inner product-preserving: for any kets $|x\rangle$ and $|y\rangle$,*

$$\langle x|y\rangle = \langle Ux|Uy\rangle.$$

Some standard single-qubit gates

We now introduce several important examples of single-qubit unitary operators. These operators act on $\mathbb{C}^2$, and therefore are represented by 2×2 unitary matrices with respect to the computational basis $\{|0\rangle, |1\rangle\}$.

This list is not exhaustive; it merely introduces some of the most commonly used gates in quantum computing.

The Pauli gates

The three **Pauli matrices** are defined as follows:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Each of these matrices is unitary (in fact, Hermitian as well).

The matrix X acts as a *bit-flip*, interchanging $|0\rangle$ and $|1\rangle$. The matrix Z acts as a *phase-flip*, introducing a relative phase of -1 on $|1\rangle$. The matrix Y combines both bit flip and phase flip effects. Indeed, it can be verified that $Y = iXZ$.

One may verify that

$$X^2 = Y^2 = Z^2 = I,$$

and therefore each Pauli matrix is its own inverse.

The Hadamard gate

The **Hadamard gate** is defined by

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Its action on the computational basis is

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \qquad H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle).$$

Thus, H transforms basis states into equal superpositions. One may verify that H is unitary and satisfies $H^2 = I$.

Phase shift gates

For any real number θ, the **phase shift gate** $P(\theta)$ is defined by

$$P(\theta) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix}.$$

Its action on basis states is

$$P(\theta)\,|0\rangle = |0\rangle\,, \qquad P(\theta)\,|1\rangle = e^{i\theta}\,|1\rangle\,.$$

This gate introduces a relative phase between $|0\rangle$ and $|1\rangle$. It is unitary for all real θ, since

$$P(\theta)^\dagger = P(-\theta), \qquad P(\theta)^\dagger P(\theta) = I.$$

Summary. The Pauli gates, the Hadamard gate, and the phase shift gates form some of the most frequently used building blocks in quantum circuits. In subsequent lectures, we will deploy them in various quantum protocols, circuits and algorithms.